

Geographer turned data analyst. Fifteen years of applied analytics, GIS, and dashboard work — at the National Park Service, a customer data platform, a geospatial-risk startup, and now as **founding technical advisor** to a small water-utility software startup. **BA in Geographic Information Science (Honors)**, certificate in Machine Learning / Data Science, MBA. Honors thesis used **ArcGIS spatial statistics** on 1,650 host-farm locations across the U.S. **Available full-time for a senior analyst role** where the GIS + Python + dashboard toolkit can support real water-conservation and demand-modeling work.

01 THE FIT — JD LINE BY LINE

BA/BS preferred in science or math	► MATCH BA, Geographic Information Science, Honors (Colgate). Plus Machine Learning / Data Science certificate (Springboard 2024) and MBA (Quantic 2021).
3+ yrs data analysis, RStudio required	► MATCH 15 yrs of applied data work. Python-first (pandas / scikit-learn / statsmodels — deeply versed). R + Snowflake on one published project (the AMI assessment) — comfortable, but honest: I'd ramp on R-specific idioms in the first weeks.
ArcGIS required , Excel + PowerPoint + Word required	► STRONG 3 yrs ArcGIS + Mapbox + Carto at NPS. Honors thesis built in ArcGIS. Excel / Word / PPT are native deliverable formats — shipping client decks + memos since 2012.
Other statistical packages required (R / SAS / SPSS / Python)	► MATCH R + Python (pandas, scikit-learn, statsmodels) + SQL. Reproducible notebooks + production pipelines both.
Power BI (preferred) / web data viz required	► STRONG Tableau, Looker, Mapbox, Carto, Observable, Plotly, D3, Recharts. Power BI = one-week ramp.
Snowflake (preferred) for client data	► ADJACENT Postgres + Supabase + BigQuery in production. Snowflake SQL + warehouse = one-week ramp.
Water demand + supply , conservation preferred	► MATCH 6 months advising a water-utility software startup w/ live Colorado pilots — compliance messaging, facility health, consumption analysis. Plus a published AMI Program Implementation Assessment (R + Snowflake) for a Colorado mountain resort utility.
Documented methods · reproducibility · comms / training	► STRONG 180-pg thesis w/ methods + appendices. README-first eng at Upstream. Wrote part of LearnOSM (HOT OSM, 23+ languages). Monthly operator trainings.

02 SHIP LOG — WATER + GIS + DATA WORK

2025 — NOW · UPSTREAM AI · FOUNDING TECHNICAL ADVISOR (PART-TIME) · BOULDER, CO

Founding advisor · early-stage *water-utility software* (pre-revenue)
Founding technical contributor to an early-stage platform serving 1-to-5-person water utilities in rural Colorado. **Two live pilots.** Built a RAG-powered document repository for utility SOPs / sampling plans / regulatory correspondence; an AI-assisted daily-operating-report workflow; compliance messaging + monthly DMR drafting; facility-health + Supabase + Recharts operator dashboards. **Pre-revenue side project — would step back to advisory for the right full-time analyst role.** Direct exposure to consumption analysis, non-revenue water, and the realities of small-utility data quality. getupstream.ai

2011 · COLGATE HONORS THESIS · GEOGRAPHY

A Network of Alternative Economies
180-pg honors thesis. Geocoded 1,650 WWOOF host farms across the continental U.S., joined to county-scale Census + USDA + ERS datasets. Multivariable regression, location quotient, **local Moran's I**, **IDW interpolation**, **kriging**. 12 field interviews ground-truthing the spatial model.

2012 — 15 · NATIONAL PARK SERVICE · FRONTEND / GIS

Team behind *NPS Park Tiles*
3 yrs in ArcGIS / Mapbox / Carto. On the **team that built NPS Park Tiles** — the basemap behind park websites used by millions. Used it for **custom park mapping projects**: wildfire perimeters, endangered-species habitat, visitor-management decision support.

2026 · SPRINGBOARD ML/DATA SCIENCE CAPSTONE (R + SNOWFLAKE)

AMI Program Implementation Assessment · Eagle River valley, CO

End-to-end AMI analysis on hourly interval data (60 meters, **525k reads**) for a Colorado mountain resort utility (Vail / Avon / Beaver Creek): seasonal demand, **anomaly detection**, customer segmentation for conservation, and **non-revenue water**. [Report ↗](#)

2009 – 10 · COLGATE · RESEARCH ASSISTANT (WATER RATES)

Conservation *water-rate adoption* research

Worked with **Prof. Manuel Teodoro** on professional mobility + water-rate structures — seed of his peer-reviewed JPART paper (*Contingent Professionalism*, 2010). Finding: executive career path predicts adoption of **conservation water rates**. Directly on-topic for MWM.

2022 – 23 · LYTICS · SR. TECHNICAL CONSULTANT

Enterprise *customer data platform*

Got enterprise clients live on a CDP — translating "we want to know our customers" into SQL queries, segments, audiences, and data pipelines that actually shipped. Heavy reporting + dashboards layer.

2018 – 21 · TERENCE · HEAD OF MARKETING TECHNOLOGY

Seed-stage *geospatial risk* startup

3 yrs at a geospatial intelligence startup. **Coordinated across engineering, product, and sales**; ran upsell + onboarding for new clients. **4x qualified leads**. Direct overlap w/ MWM's GIS scope.

03 ANALYST TOOLKIT

JD-required

Adjacent · 1-week ramp

Working knowledge

Python (preferred) · pandas · scikit-learn · RStudio · ArcGIS · MS Excel · PowerPoint · Word

SQL · Postgres · BigQuery

Power BI · Tableau · Looker

Mapbox · Carto · QGIS

Plotly · D3 · Observable

Snowflake (1-project)

Git · Notion · Linear

Anthropic Claude · OpenAI

04 RECEIPTS — WHAT MAKES THIS A REAL FIT

- ♦ **Water context by recent advisory work.** Two years as founding advisor to a pre-revenue water-utility software startup w/ live Colorado pilots. **I know SDWA, monthly reports, the texture of small-utility data, and what operators actually need from analysts.**
- ♦ **GIS by training.** 180-page honors thesis in spatial statistics (Colgate). 3 yrs of NPS cartography in ArcGIS + Mapbox + Carto. Comfortable w/ parcel + Census + utility-service-area data joins.
- ♦ **Python-first analyst** w/ a real R + Snowflake project. Machine Learning / Data Science cert (Springboard 2024). **Happy to ramp on R-specific idioms early on.**
- ♦ **Public-good orientation.** National Park Service, HOT OSM (LearnOSM curriculum in 23+ languages), rural water utilities. Fifteen years finding work that matters to communities most software companies overlook.

05 EARLIER ROLES

2022 ZAYO · 2015–18 HOMEBOT · 2011 LEARNOSM

Marketing automation · solutions engineering · open-source mapping

Zayo (2022): lead-scoring flows + dashboards across **engineering, sales, product**; ran upsell + onboarding, +30% closed-won. **Homebot (2015–18):** early-team Sol Eng Mgr — data schema for real-estate ingestion, shipped agent/lender product, upsell + implementation. **LearnOSM (2011):** HOT OSM training curriculum, 23+ languages.

06 EDUCATION

BA, GIS · *Honors*

COLGATE · 2011

Thesis: ArcGIS spatial stats on 1,650 WWOOF host farms — regression, IDW, kriging.

Machine Learning / Data Science

SPRINGBOARD · 2024

Python-first curriculum (pandas, scikit-learn, statsmodels). Statistical modeling, ML pipelines, dashboards.

MBA

QUANTIC · 2021

Strategy + finance + ops grounding to complement the technical background. Concentration in entrepreneurship.

Full-Stack Web Dev

GALVANIZE · 2014

Immersive program. Foundation for the engineering work at NPS, Homebot, Upstream.